Do Published Attention-Block Gains Survive Common Image Corruptions?
A Matched-Budget CIFAR-10 / CIFAR-10-C Stress Test of SE, BAM, and CBAM

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Abstract

Channel and spatial attention blocks are compared almost exclusively on clean test accuracy, and one of them positions itself as an improvement over another on precisely that basis. Whether the ordering this induces survives the distribution shift a deployed classifier meets has not been tested for this method family under a matched protocol. We train squeeze-and-excitation, the bottleneck attention module, and the convolutional block attention module, plus an attention-free baseline, into one shared CIFAR-scale residual backbone at an identical budget, inserting each module at the identical point so that only the module varies, and evaluate the resulting twelve checkpoints on CIFAR-10-C without retraining. The clean-accuracy ordering CBAM > SE > none > BAM is preserved at every corruption severity, on mean corruption accuracy, and on a starting-point-normalised relative degradation measure: for this method set the field's single metric was not misleading. Two secondary findings complicate the picture. Clean accuracy cannot separate the top two arms (0.11 pp against a combined across-seed spread of 0.26 pp) whereas mean corruption accuracy can (0.92 pp against 0.49 pp), so the corruption axis is here the more discriminative measurement rather than the noisier one; and the parallel channel-plus-spatial module falls 4.5 pp *below* the attention-free baseline. We also report a discrepancy between that module's published fusion equation and the fusion its official code computes.

1 Introduction

Attention blocks are among the cheapest architectural additions in modern computer vision. A handful of extra parameters inserted into an existing backbone is reported to buy a consistent accuracy gain, and squeeze-and-excitation (SE) [Hu et al., 2018], the bottleneck attention module (BAM) [Park et al., 2018], and the convolutional block attention module (CBAM) [Woo et al., 2018] have become canonical instances of the pattern [Guo et al., 2022]. Their reported gains are the reason the applied literature routinely bolts them onto convolutional networks.

Those gains are all reported on clean test data. Each of these methods is evaluated on top-1 and top-5 accuracy, and CBAM positions itself explicitly as an improvement over SE's channel-only design on that basis. None reports how the resulting models behave when the input distribution shifts.

This matters because the ordering these papers induce is what a practitioner actually consumes. Someone selecting an attention block for a deployed classifier reads off a ranking measured under conditions the deployment will not reproduce: sensors defocus, illumination drifts, contrast collapses, images are recompressed in transit. Convolutional classifiers are known to be brittle to exactly these perturbations [Hendrycks and Dietterich, 2019, Geirhos et al., 2019], and a substantial literature now treats corruption robustness as a first-class evaluation axis [Hendrycks et al., 2020]. The question that follows is therefore not whether attention blocks improve clean accuracy, but whether the *ordering* they induce on clean data is the ordering that survives corruption.

The existing literature cannot answer this, for two reasons. First, each method is validated in its own backbone, at its own scale, under its own training recipe, so a cross-paper accuracy difference conflates the module with the architecture and the optimisation schedule. Second, and more subtly, BAM and CBAM originate from a single shared public implementation and inherit a common insertion convention that an independently reimplemented SE does not; even a same-codebase comparison therefore confounds module identity with module placement.

We remove both confounds. All three attention blocks, plus an attention-free baseline, are trained into one shared CIFAR-scale residual backbone [He et al., 2016], with each module inserted at the identical point in the identical block, under an identical optimiser, schedule, augmentation policy, epoch budget and seed set. Because the module is the only term that varies, any measured difference is attributable to the module itself. We then evaluate the trained checkpoints, without retraining, on a common-corruption benchmark that shares the clean dataset's exact label space and resolution [Hendrycks and Dietterich, 2019, Krizhevsky, 2009], and re-rank the arms by mean corruption accuracy and by a relative degradation measure rather than by clean top-1.

Our hypothesis was that the ranking might not be invariant. Channel-only recalibration derives its gate from a globally pooled descriptor, whereas the spatial branches of BAM and CBAM compute attention from local feature statistics. Spatial masks depend on where informative structure lies in the image, and corruptions such as defocus blur and contrast reduction attack that structure directly, so a spatial gate that mislocalises under corruption could degrade faster than a channel-only gate. We stressed in advance that the competing outcome is equally informative, and committed to reporting it as prominently. That is the outcome we obtained.

Our contributions are:

- **A matched-budget, single-backbone comparison** of three published attention blocks against an attention-free baseline, in which backbone, insertion point, optimiser, schedule, augmentation and seed set are all held fixed, so that differences are attributable to the module rather than to its host or its placement.
- **A negative result, reported as the primary finding.** The clean-accuracy ordering is preserved at every corruption severity, on mean corruption accuracy, and on relative degradation. For this method set, ranking on clean top-1 was not misleading.
- **A measurement-power observation that qualifies it.** Clean accuracy cannot resolve the top two arms, while mean corruption accuracy can. The corruption axis is here the more discriminative measurement, which is a reason to report it even when it does not reorder anything.
- **A provenance record**, including a discrepancy between BAM's published fusion equation and the fusion its official public code computes, which to our knowledge has not previously been reported.

Section 2 positions the work. Section 3 describes the backbone, the modules and their provenance, and the fidelity checks. Section 4 gives the protocol and metrics. Section 5 reports results and answers the title question. Section 6 scopes the claim and Section 7 concludes.

2 Related Work

2.1 Channel and Spatial Attention Blocks

Squeeze-and-excitation [Hu et al., 2018] introduced channel-only feature recalibration: global average pooling reduces a feature map to a per-channel descriptor, a two-layer bottleneck multilayer perceptron maps that descriptor to per-channel gates, a sigmoid bounds them, and the features are rescaled channel-wise. The design is cheap, architecture-agnostic, and widely adopted; subsequent work has refined the channel-gating mechanism itself, for example by replacing the bottleneck with a local cross-channel interaction [Wang et al., 2020].

BAM [Park et al., 2018] and CBAM [Woo et al., 2018] extend this by adding a spatial branch. BAM computes channel and spatial attention in parallel and fuses them; CBAM applies channel attention and then spatial attention in sequence, and pools both average and maximum statistics in its channel gate. Both spatial branches derive their attention maps from local feature statistics rather than from a global descriptor, which is the property our hypothesis turned on. Broad surveys confirm the pattern across the family [Guo et al., 2022].

What unites this cluster for our purposes is its evaluation protocol. The metrics reported are clean top-1 and top-5 classification accuracy, with mean average precision added where detection is evaluated. No member of this cluster reports a robustness or distribution-shift metric, so the relative ordering these papers establish has never been tested away from the clean distribution.

2.2 Corruption Robustness Benchmarking

A parallel literature evaluates classifiers under synthetic but realistic corruptions applied at graded severities [Hendrycks and Dietterich, 2019]. The protocol fixes a set of corruption types, applies each at five severity levels, and summarises degradation across the grid, which makes it possible to compare how fast models degrade rather than only where they start. Related work established that convolutional classifiers rely on cues these corruptions disrupt [Geirhos et al., 2019], and that targeted augmentation can mitigate the damage [Hendrycks et al., 2020].

The corrupted benchmark used here shares the clean dataset's exact label space and resolution [Krizhevsky, 2009], so evaluating a trained model on it requires inference only: no retraining, no relabelling, no architectural change. That is what makes the audit cheap enough to be worth running, and makes its absence from the attention-block literature harder to explain.

Characteristically, however, this literature deploys corruption benchmarks to *validate a newly proposed intervention*. The benchmark is the evidence that a new method helps, rather than the instrument for auditing the ranking of methods that already exist.

2.3 Attention Mechanisms Under Distribution Shift

Recent work sits at the intersection. Attention-guided repair procedures improve the robustness of convolutional networks against common corruptions [Zhang et al., 2025], and channel attention has been combined with multi-scale push–pull inhibition to the same end [Ranabhat et al., 2025]. Both are validated on corruption benchmarks of the kind we use.

Both, however, are proposals: each introduces a new mechanism and uses the corruption benchmark as evidence that it works. Neither is a head-to-head comparison of the already-published blocks under a single matched training protocol with a fixed insertion point. That controlled comparison, and the ranking-stability question it makes answerable, is the gap this paper fills.

3 Method

3.1 Shared Backbone and a Single Insertion Point

All arms use one CIFAR-adapted ResNet-18: a 3×3 stride-1 stem with no initial max-pool, preserving 32×32 spatial detail, followed by four stages of two basic blocks at widths 64/128/256/512, global average pooling, and a linear classifier. Within each basic block the attention module is applied to the residual branch after the second batch normalisation [Ioffe and Szegedy, 2015] and before the residual addition:

$$\mathrm{out} = \mathrm{ReLU}\big(A\big(\mathrm{BN}_2(\mathrm{conv}_2(\mathrm{ReLU}(\mathrm{BN}_1(\mathrm{conv}_1(x)))))\big) + \mathrm{shortcut}(x)\big)\,, \tag{1}$$

where A is the attention module under test, or the identity for the baseline arm. The identity shortcut never passes through A. Figure 1 shows the block.

Because A is the only term in Equation 1 that varies, any measured difference is attributable to the module rather than to placement, host architecture, or training recipe. This is the study's primary confound control. It has a cost, which we state plainly: no arm reproduces its own paper's placement — official BAM sits *between* stages rather than inside blocks — so absolute accuracies here are not comparable to published numbers, and we claim none. Only the within-study ordering is claimed.

The arms differ only by attention parameters: 11.17M for the baseline, 11.26M for SE, 11.36M for BAM, and 11.26M for CBAM.

3.2 Attention Modules and Their Provenance

SE is reimplemented from Equations 2–3 of Hu et al. [2018]: global average pool to $z \in \mathbb{R}^C$, then $s = \sigma(W_2\,\delta(W_1 z))$ with $W_1 \in \mathbb{R}^{(C/r)\times C}$, then channel-wise rescale by s. This is the study's one reimplementation, and it is necessary:

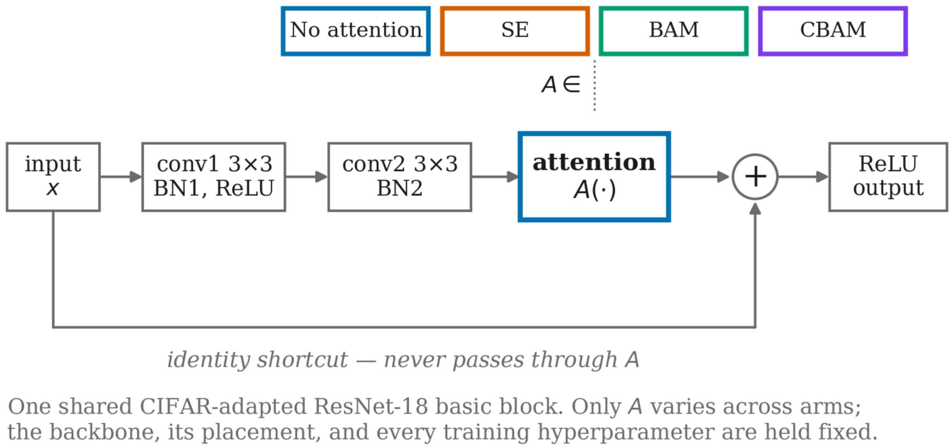

Figure 1: The shared residual block and its single attention slot. Every arm uses this identical block; only the module A changes, and the identity shortcut never passes through it. Holding the insertion point fixed is what makes a measured difference attributable to the module rather than to its placement.

the official release is Caffe-only, consisting of `.prototxt` model definitions with custom `.cpp`/`.cu` layers, and contains no Python. The block is fully specified by two equations, so the reimplementation is unambiguous. We use $r = 16$, the paper's default, and record it as one point in the design space rather than a re-derivation of a tuned value.

BAM and **CBAM** are ported directly from the module definitions of the official public implementation. We take only the module definitions; the repository's own ImageNet-scale model file is not used. Layer types, ordering, pooling choices, bias settings, batch-norm placement and hyperparameters, C/r bottleneck widths, channel-pool concatenation order and BAM's dilation-4 convolution stack are all preserved. The only change is replacing a deprecated sigmoid entry point removed from modern versions of the framework; the repository has had no commits since March 2023 and ships no dependency pins.

Two details are load-bearing, because conflating them would silently weaken the comparison.

BAM's and CBAM's channel gates are different mechanisms. Both files define a class of the same name, but BAM's pools average statistics only and places a one-dimensional batch normalisation between its MLP layers, while CBAM's pools average *and* maximum statistics through a shared MLP with no normalisation. Sharing one implementation between the two arms would partially collapse the channel-only-versus-channel-plus-spatial contrast this study measures.

BAM's paper and its official code disagree on the fusion operator. Park et al. [2018] specify additive fusion, $M(F) = \sigma(M_c(F)+M_s(F))$. The official code instead computes $\text{att} = 1+\sigma(M_c(F)\cdot M_s(F))$ and returns $\text{att}\cdot F$: a multiplicative fusion with a residual offset. Since the artefact this literature actually reuses is the code rather than the equation, we follow the code, and we report the discrepancy rather than resolving it silently. To our knowledge it has not previously been noted.

3.3 Mechanical Verification of Module Fidelity

The comparison's validity rests entirely on the modules being what they claim to be, so fidelity is asserted mechanically rather than by inspection. A verification suite of 54 checks asserts structural properties — BAM's average-only pooling and one-dimensional batch normalisation, CBAM's dual pooling and batch-normalised 7×7 spatial convolution, SE's pooling, bottleneck width and sigmoid gate — and, critically, behavioural ones. BAM's output-to-input ratio must lie in $(1, 2)$, as $1 + \sigma(\cdot)$ requires, and CBAM's in $(0, 1)$, as a pure multiplicative gate requires. Fusion semantics are thus verified by behaviour rather than by reading source code. The suite also asserts that the two

Table 1: Top-1 accuracy (%) by corruption severity, averaged over four corruption types and three seeds, with across-seed standard deviation. mCA is the mean over severities 1–5; relative drop is (clean − mCA)/clean, where lower is better.

Model	Clean	Sev. 1	Sev. 2	Sev. 3	Sev. 4	Sev. 5	mCA	Rel. drop
none	86.09±2.10	83.96±2.25	80.99±2.93	75.53±3.00	67.12±2.84	55.01±0.99	72.52±2.31	15.77%
SE	89.21±0.18	87.29±0.10	85.05±0.05	80.09±0.23	71.98±0.06	57.16±0.05	76.31±0.04	14.45%
BAM	81.56±2.47	79.28±2.71	75.51±3.09	70.08±3.08	62.83±2.67	52.91±2.20	68.12±2.54	16.49%
CBAM	**89.32**±0.08	**87.68**±0.22	**85.58**±0.28	**80.94**±0.67	**73.13**±0.49	**58.87**±0.77	**77.24**±0.45	**13.52%**

channel-gate classes are distinct objects, so the defect described in Section 3.2 cannot silently return. It runs before every training installment.

4 Experimental Setup

4.1 Data and Training Protocol

Training and clean evaluation use CIFAR-10 [Krizhevsky, 2009]: 50,000 training and 10,000 test images at 32×32 across 10 classes. Corrupted evaluation uses CIFAR-10-C [Hendrycks and Dietterich, 2019], restricted to four corruption types — brightness, contrast, defocus blur and elastic transform — at severities 1 through 5, with 10,000 images per type and severity. Test-time preprocessing is identical for clean and corrupted images.

Every arm receives an identical recipe: SGD with learning rate 0.1, momentum 0.9, weight decay 5×10^{-4}, batch size 128, cosine annealing [Loshchilov and Hutter, 2017], and standard augmentation (random 32×32 crop with 4-pixel padding, random horizontal flip). The budget is 10 epochs per run and the seed set is $\{1, 2, 3\}$, giving 12 training runs. No arm uses a pretrained or externally obtained checkpoint. Mixed precision [Micikevicius et al., 2018] is used for training; validation runs in full precision so that checkpoint selection and final evaluation share one numeric path. All runs used a single NVIDIA RTX A2000 with 6 GB of memory.

4.2 Metrics

We report three. *Clean top-1 accuracy* is the metric the field currently ranks on and the severity-zero reference point. *Mean corruption accuracy* (mCA) at each severity averages over corruption types. *Relative robustness drop*, (clean − corrupted)/clean, normalises out each arm's clean starting point so that an arm beginning lower is not mistaken for a robust one merely because it had less accuracy to lose.

All results are averaged over seeds and reported with the across-seed standard deviation. Before any ordering is claimed, we check whether the gap between the top two arms exceeds their combined across-seed spread. We compare against the spread of the two arms being compared, not the largest spread anywhere in the grid; the latter belongs to the higher-variance baseline and BAM arms and would wrongly declare a resolvable difference unresolved.

5 Results and Analysis

5.1 Accuracy Across Corruption Severity

Table 1 reports every arm at every severity, with mCA and relative drop. Figure 2 plots the same data.

Absolute accuracy falls steeply with severity for every arm, from roughly 86 to 89% clean down to 53 to 59% at severity 5. The arms do not, however, change places.

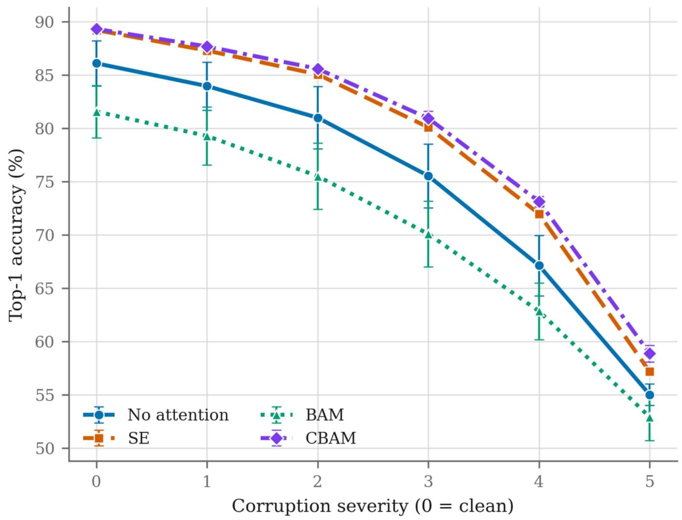

Figure 2: Top-1 accuracy against corruption severity, averaged over four corruption types and three seeds, with across-seed standard deviation as error bars. The curves do not cross at any severity: the ordering established on clean data is the ordering at every level of corruption.

Table 2: Ranking of the four arms at each condition. No reordering occurs anywhere.

Condition	Ranking (best first)	Same as clean?
Clean (severity 0)	CBAM > SE > none > BAM	—
Severity 1	CBAM > SE > none > BAM	yes
Severity 2	CBAM > SE > none > BAM	yes
Severity 3	CBAM > SE > none > BAM	yes
Severity 4	CBAM > SE > none > BAM	yes
Severity 5	CBAM > SE > none > BAM	yes
mCA (severities 1–5)	CBAM > SE > none > BAM	yes
Relative robustness drop	CBAM > SE > none > BAM	yes

5.2 Does the Clean Ranking Survive?

Table 2 answers the title question directly.

The clean-accuracy ranking survives. It is preserved at all five severities, on mean corruption accuracy, and on relative robustness drop. Figure 3 shows the same result as rank trajectories: a reordering would appear as crossing lines, and none occurs. This is the null outcome we pre-committed to reporting, and for this method set it means the field's single clean-accuracy metric was not a misleading basis for choosing among these blocks.

The relative-drop column is worth isolating, because it is the measure designed to catch a spurious robustness advantage. CBAM loses 13.52% of its clean accuracy, SE 14.45%, the baseline 15.77%, and BAM 16.49%. The ordering is unchanged, so CBAM's advantage under corruption is not an artefact of its higher starting point: it degrades more slowly in proportional terms as well as finishing higher.

This runs against our stated hypothesis. We expected the spatial gates of BAM and CBAM to be more fragile than SE's global channel gate under blur and contrast reduction. CBAM, which has a spatial gate, in fact degrades the most slowly of all four arms. We report the hypothesis as refuted rather than quietly restating it.

5.3 What the Two Metrics Can and Cannot Resolve

A ranking is only meaningful if its adjacent pairs are separated by more than seed noise, and here the two metrics differ in what they can resolve.

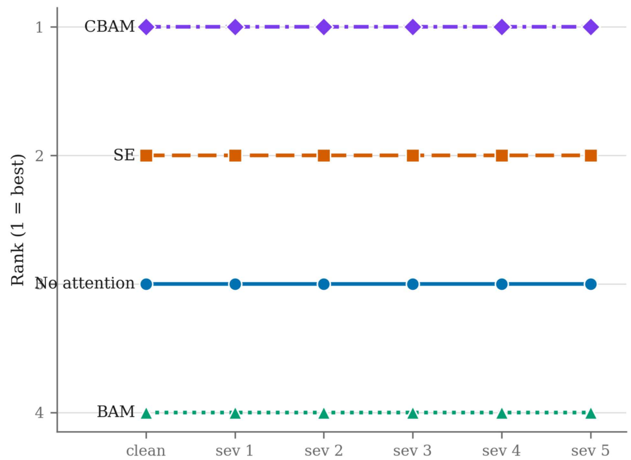

Figure 3: Rank position of each arm at clean and at each corruption severity. A reordering would appear as crossing lines; none occurs, which is the paper's central negative result in its most direct form.

Table 3: Mean accuracy (%) per corruption type, averaged over severities 1–5 and three seeds.

Model	Brightness	Contrast	Defocus blur	Elastic transform
none	83.76	59.66	72.28	74.38
SE	86.83	67.06	74.63	76.75
BAM	78.66	53.27	69.40	71.16
CBAM	**87.33**	**67.53**	**76.05**	**78.05**

On clean accuracy the top two arms are 0.11 pp apart, against a combined across-seed standard deviation of 0.26 pp. **Clean accuracy cannot distinguish CBAM from SE at this budget.** On mean corruption accuracy the same two arms are 0.92 pp apart against a combined spread of 0.49 pp, which does exceed seed noise.

The corruption measurement therefore resolves a difference between the two best arms that the clean measurement cannot. This is a distinct point from the ranking result and cuts the other way: the corruption axis did not reorder anything, but it discriminated more finely than the metric the field already reports. A practitioner choosing between these two blocks on clean top-1 would be choosing on a coin flip; the same choice on mCA is supported by the data.

The larger spreads in the table belong to the baseline and BAM arms ($\pm$2.10 and $\pm$2.47 pp clean, against $\pm$0.18 and $\pm$0.08 pp for SE and CBAM). Attention appears to stabilise training across seeds at this budget as well as improving the mean, though with three seeds we report this as an observation rather than a claim.

5.4 Per-Corruption Breakdown

Averaging four corruption types can hide a type-specific skew, so Table 3 reports the breakdown.

The ordering is identical within every corruption type, so the aggregate is not concealing a reversal. Contrast is by far the most damaging corruption, costing the baseline 26 pp relative to brightness, and it is also where the attention arms separate most from the baseline: CBAM leads the baseline by 7.9 pp on contrast against 3.6 pp on brightness. A global channel gate can rescale channels whose activations have been uniformly compressed, which is a plausible mechanism for that pattern, but our design does not isolate it and we offer it as a hypothesis for future work rather than a finding.

5.5 BAM Underperforms the Attention-Free Baseline

BAM finishes last on every metric, 4.5 pp below the baseline on clean accuracy and 4.4 pp below on mCA, with the widest across-seed spread of any arm, and the largest relative degradation of the four (Table 1).

We are deliberately cautious about what this means. At a 10-epoch budget, an arm that converges more slowly is indistinguishable from an arm that converges to a worse optimum, and BAM is the heaviest module here, with dilated convolutions and an extra batch-normalised MLP in every one of the eight blocks. BAM's mean final-epoch training accuracy (82.79%) is also the lowest of the four arms, against 92.48% for CBAM and 87.90% for the baseline, which is consistent with underfitting rather than with a robustness deficit. We therefore report that *BAM did not pay for itself at this budget*, and explicitly do not claim that BAM is a worse module than the baseline in general. Section 6 states what would settle it.

6 Limitations

Training budget. Ten epochs is matched across arms, which is what the ranking claim requires, but it is short. Absolute accuracies would rise with a longer schedule, and the ordering could in principle change — most plausibly for BAM, whose deficit (Section 5.5) is consistent with underfitting. The single most valuable extension is to rerun the identical grid at a longer budget and check whether BAM's deficit closes.

Three methods and one baseline. The design supports per-severity descriptive statements about whether the ranking held. It does not support a general claim that attention *type* predicts corruption robustness; four points do not characterise a design space.

Four of the benchmark's corruption types. Averaging over brightness, contrast, defocus blur and elastic transform could flatter or punish spatial attention relative to the full benchmark. We report the per-type breakdown alongside the mean so a skew is visible, and scope the finding to these four types. All types in the source are ungated, so extension is inference-only and requires no retraining.

Insertion point. Holding the slot fixed is what buys attribution to the module, but it means no arm reproduces its own paper's placement. Absolute accuracies are therefore not comparable to published results and none is claimed.

Three seeds, one dataset, one resolution. Three seeds is enough to show that the top-two clean gap is inside seed noise, but not to place tight intervals on the arms with wider spread. Results are CIFAR-scale only.

7 Conclusion

We trained three published attention blocks and an attention-free baseline into one shared backbone, at one insertion point, under one budget and one seed set, and re-ranked them under common image corruptions rather than on clean accuracy alone.

The ranking survived. CBAM > SE > none > BAM holds at every corruption severity, on mean corruption accuracy, and on relative degradation. For this method set, the clean-accuracy metric the literature reports was not a misleading basis for choosing between these blocks, and our hypothesis that spatial gates would prove more fragile than a global channel gate was refuted: the arm with a spatial gate degraded most slowly.

Two findings qualify that. Clean accuracy could not resolve the top two arms while mean corruption accuracy could, so the corruption axis was here the more discriminative measurement even though it reordered nothing — a reason to report it that does not depend on it overturning anything. And BAM did not pay for itself at this budget, finishing below the attention-free baseline, though at ten epochs we cannot separate a worse optimum from slower convergence.

We also record that BAM's official implementation computes a multiplicative gate fusion where its paper specifies an additive one. An implementation this widely reused differing from its own paper on the fusion operator is worth the field's attention independently of our results.

The cheapest extensions are clear: the remaining eleven corruption types are ungated and require inference only, and rerunning the identical grid at a longer budget would settle whether BAM's deficit is underfitting.

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